

研究自述

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[English version]

環境品質取決於誘因。我的研究探討誘因變動如何改變資源使用、環境品質與社會福利。我關注個人與企業決策所產生、但市場無法充分內部化的環境外部性，研究場域涵蓋農業、林業、交通運輸與能源。

我的研究包含兩個相互連結的主題。第一，探討資源依賴型家戶與企業如何在誘因變動下管理環境資源；第二，探討交通系統如何產生外部性，以及政策與外部衝擊如何改變旅運行為。兩者共同檢視經濟決策如何與環境限制及公共政策互動。

1. 環境與自然資源管理

農業與林業仰賴自然資源，但其環境價值往往未反映於市場價格。本研究探討資源依賴型家戶如何回應誘因變化，以及這些反應如何影響生產、環境結果與風險。

- 農家經濟 Liou et al. (2025a)、Lee et al. (2025a)、Chang and Liou (2026) 與 Lee et al. (2025b) 探討農業與森林家戶如何進行生產決策、管理環境資源並承擔經濟風險。再生能源採用可提高農場銷售與勞動需求，與農業生產形成互補。食品溯源制度可降低化學肥料、農藥與地下水使用，同時提高收入與市場管道。在林業方面，較高的樹種多樣性雖降低預期收入，卻能降低收入波動，顯示生物多樣性是兼具風險管理功能的自然資本。
- 氣候政策 Liou et al. (2025b) 將研究延伸至國際氣候政策。該研究以理論模型分析碳邊境調整政策如何改變企業採用綠色技術的誘因，並重塑全球生產決策。

2. 交通運輸與環境外部性

交通系統透過交通事故、壅塞、空氣污染與碳排放產生重要外部性。本研究探討交通政策與大型衝擊如何影響旅運行為、環境結果與社會福利。

- 大型事件與道路安全 Liou and Chang (2026a)、Liou and Chang (2026b) 與 Chang et al. (2024) 探討制度變革與大型事件如何影響交通風險與環境結果。交通規則改革可降低事故、傷害與醫療成本；大型活動則主要透過改變交通需求，提高交通風險與污染。

- 旅運需求與 COVID-19 Liou et al. (2025c)、Chang et al. (2021)、Liou et al. (2024) 與 Liou et al. (2026) 探討交通系統與地方經濟如何回應政策變化與大型外部衝擊。大眾運輸定價可提升移動性並降低碳排放；COVID-19 則透過改變風險認知與消費決策，重塑旅運行為並延緩經濟復甦。

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